

ARINDAM SHARMA

MS Automotive and Motorsports | Vehicle Dynamics and Control

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Professional Profile

Masters student in Automotive Engineering & Motorsports at Politecnico di Milano specialising in control systems, data analysis, vehicle dynamics and dynamic system simulation, with four years of Formula Student engineering experience. Experienced in MATLAB/Simulink and Python for vehicle modelling, controller design and tuning, telemetry analysis, and data-driven vehicle performance optimisation. Hands-on experience in suspension kinematics, drivetrain design, vehicle setup, on-track testing, and correlation of simulation with test data across Formula Bharat, FS East, and FS Germany. Developed longitudinal, lateral, ride, and multibody vehicle models, with a focus on translating simulation and test data into vehicle performance improvements. Utilises LLM-assisted development to accelerate technical workflows, debugging, and exploration of simulation methodologies. Former Team Captain of Pravega Racing, leading a 30-member engineering team through vehicle development and competition.

Education

MS in Automotive & Motorsports | Politecnico di Milano

Sept 2025- Ongoing

Grade : 24.78 /30

Relevant Coursework - Vehicle Dynamics and Control, Data Analysis, Control of Mechanical Systems

B.Tech in Mechanical Engineering | VIT Vellore

July 2021- July 2025

Grade : 9.20/10

Relevant Coursework - Kinematics of Machines, Probability and Statistics, Engineering Physics, Engineering Optimisation

Experience

Team Captain | Formula Student

Pravega Racing

Dec 2023-Dec 2024

- Led a 30-member engineering team, coordinating cross-subsystem vehicle integration and engineering deliverables.
- Directed on-track testing and suspension setup optimisation for Formula Bharat 2025 using structured **data-driven tuning, telemetry analysis, and driver feedback correlation**.
- Achieved **5 kg driveline mass reduction** and **3 hp drivetrain performance gain** through sprocket geometry and differential mounting redesign, validated against baseline simulation.
- Authored the **Structural Equivalency Spreadsheet (SES)** and **Engineering Design Report** with meticulous accuracy and led Engineering Design and Cost Report events, contributing to 1st Overall in both at Formula Bharat 2025.

Drivetrain & Performance Engineer | Formula Student

Pravega Racing

Apr 2022-Jan 2025

- Contributed to scrutineering, vehicle setup, on-track testing, and real-time data analysis.
- Developed **MATLAB/Simulink gear-ratio optimisation** algorithms for longitudinal acceleration, achieving a time of **4.9s** for the Acceleration event later validated against track runs while working with Race Dynamics datalogging system.
- Assisted with launch-control implementation and **ECU calibration**, to optimise air-fuel mixture composition and engine response.
- Executed bench testing of the powertrain along with **differential tuning** through ramp angles and preload to maximise traction.
- Contributed across multiple engineering domains during the manufacturing and assembly of two FSAE cars, supporting drivetrain, suspension, steering, electrical, and vehicle integration activities.

Systems Engineering Intern

Larsen and Toubro-PES

Jan 2025-Apr 2025

- Developed full CAD assemblies in Siemens NX for precision mechanical systems and performed analytical load calculations and ANSYS FEA to validate designs against operational load cases. Maintained BOM and design reports for all components.
- Automated load and structural calculations using Python, improving analysis efficiency and reducing repetitive manual calculations; worked to defence-grade tolerance and documentation standards.

Key Projects

<https://github.com/arindam-30/Automotive>

Vehicle Dynamics Simulation Models

2026

- Developed Pacejka MF5.2 tyre models in MATLAB using TTC Hoosier tyre data, fitting pure and combined-slip characteristics across multiple vertical-load conditions.
- Built a 3-DOF longitudinal vehicle dynamics model in MATLAB/Simulink incorporating vehicle body motion, wheel rotational dynamics, drivetrain inertia, and Pacejka-based tyre forces.
- Implemented a **PI cruise controller** and an **ABS controller** on the longitudinal model, using wheel slip from the Pacejka tyre forces. The cruise controller was tuned for stable speed tracking under varying road gradients and vehicle loads.
- Developed a 4-wheel dual-track lateral dynamics model from first principles to analyse yaw rate, body sideslip, and lateral acceleration under open-loop steering manoeuvres, accounting for nonlinear tyre forces and load-transfer effects.
- Implemented a **torque vectoring controller** on the lateral model and studied its effect on yaw rate response and vehicle stability.
- Developed a 2-DOF quarter-car ride model to evaluate suspension dynamics, and ride response using frequency-domain analysis and parameter sensitivity studies. Implemented a **skyhook control strategy** to compare active and semi-active ride control.
- Optimised double-wishbone pushrod suspension kinematics by analysing camber gain, roll-centre height, anti-dive, and anti-squat characteristics against target handling requirements in MSC Adams.

- Processed IMU and GPS telemetry in MATLAB using **Butterworth low-pass filtering** and **signal conditioning** to extract reliable vehicle dynamics data from track tests.
- Extracted vehicle speed, steering corrections, peak lateral and longitudinal acceleration, braking performance, and transient response metrics to evaluate vehicle behaviour across test runs.
- Reconstructed **vehicle trajectories from GPS data** and identified key events such as peak-speed locations, steering corrections, braking zones, and acceleration phases.
- Developed MATLAB scripts to **automate telemetry filtering, metric extraction, trajectory visualisation**, and run-to-run comparison for repeatable vehicle performance analysis.

- Modelled dynamic systems in MATLAB as **transfer functions and state-space models**, and checked open-loop stability using pole locations and eigenvalues.
- Designed PI/PD controllers and verified closed-loop stability using Nyquist plots, root locus, and pole and eigenvalue analysis.
- Implemented and tuned the controllers in Simulink, using **closed-loop simulations** to check that the results matched the MATLAB analysis.
- Tuned proportional and integral gains to **minimise steady-state error** and reduce settling times, while keeping transient response acceptable.
- Modelled a **PI-based electromechanical speed control** loop with motor and mechanical dynamics, and evaluated it using step response, steady-state error, DC gain, and bandwidth.

- Designed the rear driveline architecture for a single-motor FSAE EV, covering shaft geometry, bearing selection, differential integration, and packaging constraints.
- Developed a parameterised MATLAB/Simulink **battery model** for cell-level discharge, SOC estimation, and thermal response under FSAE acceleration and endurance duty cycles.
- Performed battery thermal-management simulations to compare air and liquid cooling architectures and determine cooling requirements.
- Developed a parameterised vehicle simulation framework to streamline design iterations and enable correlation with measured on-car test data.

- Developing a **parametric double-wishbone suspension model** in Simscape Multibody for suspension geometry optimisation and kinematic analysis.
- Integrating the suspension multibody assembly into a full-vehicle model for system-level vehicle dynamics simulation and virtual testing.
- Explored the use of **MATLAB tools** such as **Virtual Vehicle Composer** for vehicle modelling to take a reference and create Reduced-Order Models for vehicle performance analysis.
- Architecting an extensible vehicle simulation framework targeting a low-cost Driver-in-the-Loop (DiL) simulator with real-time driver interaction, hardware-in-the-loop integration, and scenario-based testing.

Skills

- MATLAB
 - Simulink
 - Python
 - Model-Based Design
 - CAD- Siemens NX , Solidworks
 - Ansys - Mechanical , Fluent
 - CFD
 - OpenFOAM
 - MSC Adams
 - VIGrade
 - Data Analysis
 - MoTeC i2
- Control System Design
 - Stability Analysis
 - PID Controller Tuning
 - Technical Documentation
 - Git / Github
 - Vehicle Dynamics

Achievements

- 3rd Overall – Formula Bharat 2025.
- 1st Overall – Engineering Design & Cost Report, Formula Bharat 2025.
- 1st – Business Plan Presentation, Formula Bharat 2023.
- 2nd – Business Plan Presentation, Formula Student East 2023.
- Top 10 – All Static Events, Formula Student Germany 2023.
- Merit Scholarship – Year 1, VIT Vellore (top academic rank in cohort).